

Edge Intelligence MACSE604

Lab 6 : Video Classification - 19th Feb 2026

Insights generated

1. Used a pre-trained CNN model (MobileNetV2) for frame-level image classification.
2. Learned how `cv2.VideoCapture()` (cap) is essential for opening, reading, and controlling video streams frame by frame.
3. How to access video metadata such as frame count, resolution, and FPS using `cap.get()` properties.
4. Importance of video resolution (width × height) in determining visual quality and computational cost.
5. Common resolution standards: 640×480 (480p – SD), 1280×720 (720p – HD), 1920×1080 (1080p – Full HD), 2560×1440 (1440p – 2K Quad HD), and 3840×2160 (4K).
6. Higher resolution increases clarity but also increases memory usage and processing time.
7. Frame rate (FPS – Frames Per Second) determines how smooth a video appears and how many frames must be processed per second.
8. The difference between actual video FPS and label FPS in annotated datasets; mismatches can cause incorrect frame–label alignment.
9. The role of annotations in video files for supervised learning tasks such as detection and tracking.
10. OpenCV loads frames in BGR format, while deep learning models typically expect RGB format.
11. Only BGR and RGB channel orders are commonly used in image processing pipelines.
12. Swapping BGR and RGB channels changes color interpretation (e.g., red regions appear blue), affecting model predictions.
13. Observed how incorrect preprocessing can silently degrade classification accuracy even if the code runs.
14. Video processing requires balancing quality and transmission efficiency.

15. Recognized that telecom systems (e.g., fiber optics) aim for high-speed transmission close to the speed of light, but bandwidth and infrastructure still limit practical performance.